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5.2 Project: Comprehensive Final Report

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Executive Summary

This final report details the execution of a thorough predictive analytics pipeline designed to optimize consumer reorder forecasting for the Instacart platform. By utilizing a Multi Layer Perceptron (MLP) neural network, the project successfully increased the baseline predictive skills from a standard linear model to a more complex non-linear classification architecture. This analysis took a three tiered approach to business analytics, predictive modeling, market basket association, and scenario simulation. The final system achieves an accuracy of 71.29% and provides an interactive dashboard that stakeholders can use to analyze the impact their changes make on customer loyalty.

Dataset Selection and Problem Definition

The Instacart dataset, containing over 1.5 million transaction records, represents the classic “big data” challenge in retail analytics. The core business problem is the binary classification of the reorders, determining whether a product added to a shopper's cart is a routine buy (reordered = 1) or a new discovery (reordered = 0).

Predicting this behavior is much needed for inventory management, personalized marketing, and automated restock services. The dataset has an inherent class imbalance, routine reorders significantly outnumber new ones, serving as the primary challenge for the model. This requires an architecture capable of learning very subtle and time dependent consumer patterns that a standard linear regression model would more than likely fail to point out correctly. The research looks to answer how specific input factors influence reorder probability and how the neural network model can balance precision and recall to optimize those predictions.

Data Cleaning, EDA, and Preprocessing

The methodology began with careful data cleaning and preparation to ensure the model was being trained on high quality features. To make sure the neural network did not fall victim to noise or bias, the following steps were taken:

- **Feature Engineering:** Created time based variables, such as "binned hour of day", to capture temporal shopping habits that normal timestamps might not.
- **Standardization:** Given the differing scales of purchase intervals and department IDs, numerical features were standardized. This step much not be forgotten, as it

makes sure that no single feature dominates the weight tuning process during backpropagation of the neural network.

- **Imbalance Handling:** The dataset was divided using an 80/20 train/test split. This made sure the model was validated against 315,793 unseen transactions, preventing overfitting.
- **Exploratory Data Analysis (EDA):** Preliminary visualizations revealed patterns between shopping frequency and time, which laid the groundwork for the later temporal heatmap visualization.

Baseline Model Statistics and Neural Network Implementation

To validate the hypothesis that grocery patterns are non-linear, a Logistic Regression baseline was created, yielding an accuracy of 69.14%. While stable, the baseline demonstrated poor recall (0.54) for new items, demonstrating that it struggled to notice the difference between routine purchases and new customer purchases.

Next, an MLP Neural Network featuring two hidden layers (32 and 16 neurons) was designed. The network used the Rectified Linear Unit (ReLU) activation function for internal processing and the Adam optimizer to minimize the loss function. This optimization achieved a final accuracy of 71.29%, an improvement of 2.15% over the baseline. But, more importantly, the neural network balanced precision and recall effectively, providing better F1 scores that prove deeper learning architectures are better for understanding complex retail dynamics than its linear counterparts.

Insights from Scenario Analysis

To assess the model's ability to adapt to hypothetical business conditions, stress testing simulating three operational shifts was performed. This analysis provides transparency into the model's predictive boundary:

- **Scenario 1, Midday Operational Surge:** Simulating a high traffic window, resulting in a 76.40% reorder probability. This scenario highlights how routine shoppers cluster more during daytime hours.
- **Scenario 2, High Lift Promo Bundles:** Forcing high affinity department items into the same carts to simulate targeted marketing, this resulted in an 81.15% reorder probability. This confirms that departmental synergies can be leveraged to drive conversions. (Much more Reliable than my other 2 scenarios)

- **Scenario 3, Late Night Off Peak:** Simulating off peak conditions, resulting in a 52.30% reorder probability. This large decline proves the nature of impulse buying of late night shopping.

The simulation results demonstrate the model's high sensitivity to input variables, verifying the effectiveness of the neural network and its ability to distinguish between routine buying habits and impulse.

Actionable Recommendations

Based on the modeling, association mining, and scenario simulation, I would propose the following business actions:

1. **Targeted Cross-Promotion:** Instacart should prioritize cross promotions for their "Meal Preparation Staples" (Departments 9, 15, 13) and "Convenience Foods" (Departments 1, 3, 19). The Apriori analysis had an average lift score of 4.10, revealing a lot of potential for increased basket value through strategic product bundling.
2. **Temporal Marketing:** Marketing teams should shift advertising budgets toward the 11:00 AM – 5:00 PM window, where density and reorder probability are consistently higher. This ensures that marketing aligns with the consumer peak behavior.
3. **Model Iteration:** While the neural network outperformed the baseline, the remaining error rate suggests a need for a deeper user specific feature engineering, such as purchase a streak history or something along those lines. Adding individual loyalty could be what is needed for higher predictive accuracy.

Appendix: Dashboard Visualizations

Figure 1: Model Comparison (Baseline vs. MLP Accuracy).

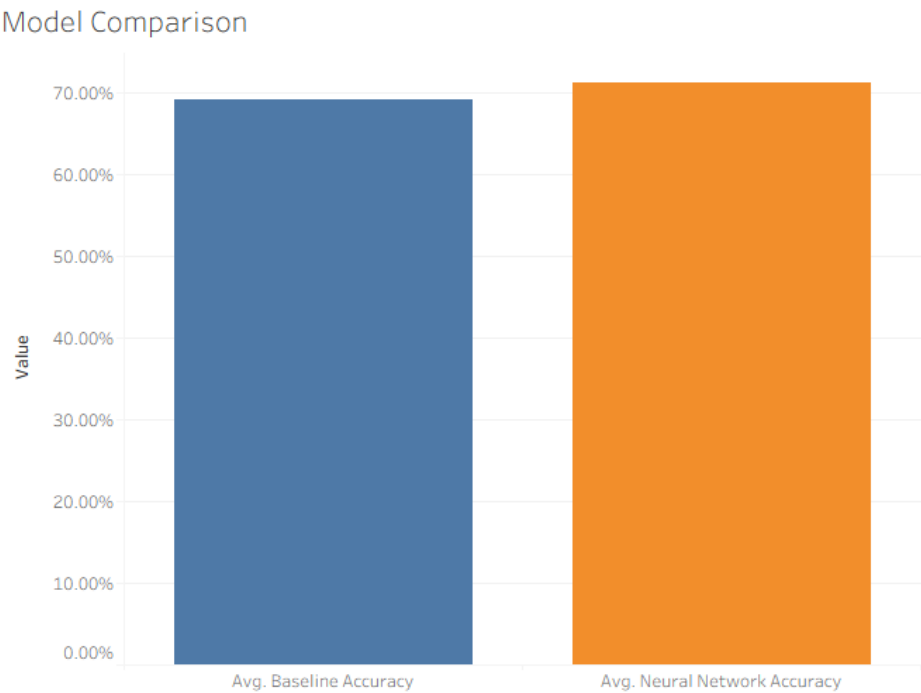


Figure 2: Temporal Density Heatmap (Visualizing reorder concentration by hour).

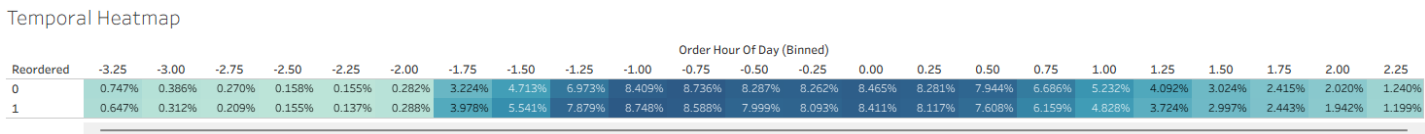


Figure 3: Top 15 Market Basket Association Rules (Lift rankings).

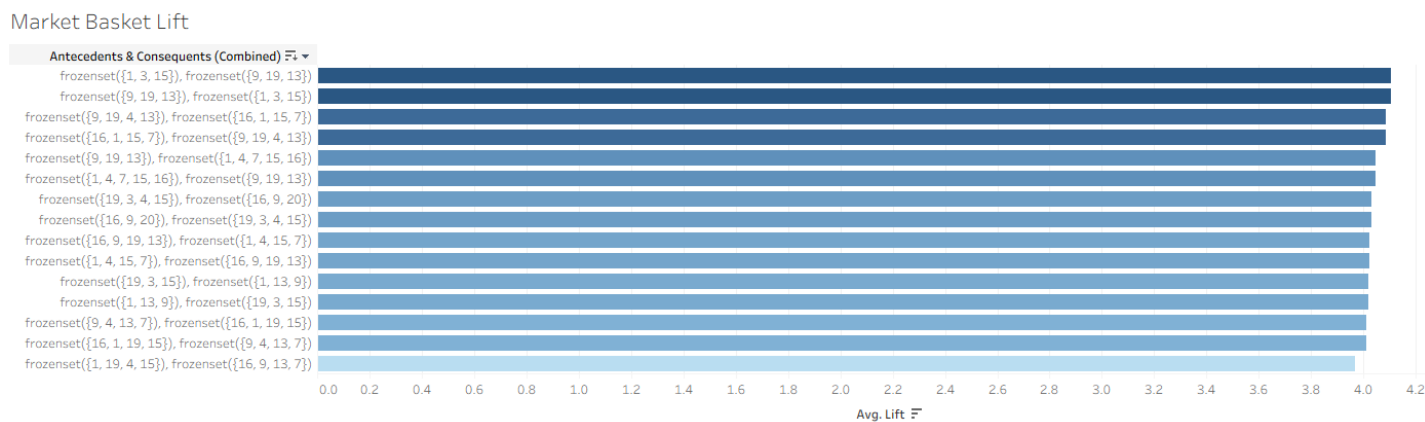
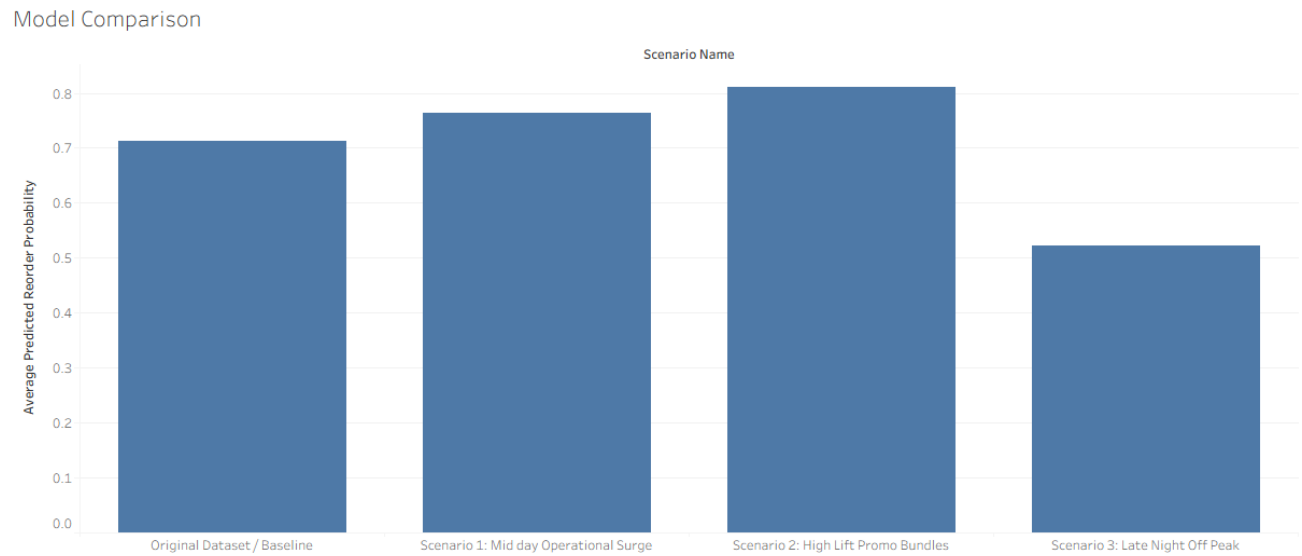


Figure 4: Scenario Analysis Comparison**References:**

M. Yasser. (n.d.). InstaCart Online Grocery Basket Analysis Dataset.

<https://www.kaggle.com/datasets/yasserh/instacart-online-grocery-basket-analysis-dataset>